🛡️ Phishing Email Detection & Awareness System

> A cybersecurity project focused on analyzing phishing emails, identifying threat indicators, and building user awareness to reduce social engineering risks.

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Overview

Phishing is one of the most common and dangerous cyber threats faced by individuals and organizations today. Attackers impersonate legitimate entities to trick users into revealing sensitive information such as passwords, banking details, or OTPs.

This project focuses on:

- Analyzing real-world phishing email samples
- Identifying indicators of compromise
- Classifying email risk levels
- Educating users with practical prevention guidelines

Objectives

#	Goal
1	Analyze real phishing email samples

#	Goal
2	Identify common phishing indicators
3	Classify emails as **Safe**, **Suspicious**, or **Phishing**
4	Explain phishing attacks in non-technical language
5	Create actionable awareness guidelines for users

Phishing Attack Types

- **Email Phishing** — Mass deceptive emails sent to large audiences
- **Spear Phishing** — Targeted attacks against specific individuals
- **Clone Phishing** — Duplicates of legitimate emails with malicious links
- **Whaling** — High-value targets such as executives or CEOs

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Methodology

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1. Collect phishing email samples from public datasets
2. Analyze email headers using specialized tools
3. Inspect sender domains and embedded URLs
4. Identify phishing indicators
5. Classify email risk level
6. Document findings
7. Create awareness guidelines
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Tools Used

Tool	Purpose
Kaggle / SpamAssassin	Public phishing email datasets
Google Admin Toolbox	Email header analysis
MXToolbox	Email header & DNS analysis
Browser DevTools	URL inspection & hover analysis
MS Word / PDF	Documentation

Email Analysis Examples

1. 🟥 PayPal Spoofing Email

Field	Value
Subject	Your PayPal Account Has Been Limited
Sender	service@paypa**1**.com
Indicators	Domain uses `1` instead of `l`, urgency tactic, fake login link
Classification	🔴 **Phishing**

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2. 🟥 Office 365 Credential Attack

Field	Value
Subject	Password Expiry Notice
Sender	admin@office365-security.com
Indicators	Fake Microsoft domain, urgency ("expires today"), redirects to phishing page
Classification	🔴 **Phishing**

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3. 🟥 DHL Delivery Scam

Field	Value
Subject	Delivery Failed
Sender	support@dhl-delivery.net
Indicators	Unexpected email, malware attachment, fake domain
Classification	🔴 **Phishing**

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4. 🟡 Bank Alert Scam

Field	Value
Subject	Your Account Will Be Blocked
Sender	security@bank-alerts.net
Indicators	Threatening tone, requests login verification, suspicious link
Classification	🟡 **Suspicious / Phishing**

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5. 🟥 Lottery Scam

Field	Value
Subject	You Won ₹10,00,000
Sender	lottery@winnerpromo.xyz
Indicators	Too-good-to-be-true offer, requests personal information
Classification	🔴 **Phishing**

Phishing Indicators

Watch out for these red flags in any email:

- 🚩 Spoofed or fake sender address
- 🚩 Suspicious or shortened URLs
- 🚩 Urgent or threatening language
- 🚩 Unexpected attachments
- 🚩 Poor grammar and spelling
- 🚩 Requests for sensitive information (passwords, OTPs, card numbers)

Email Classification

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┌─────────────┬─────────────────────────────────────────┐
│  SAFE  ✅    │ Verified sender, no suspicious links,   │
│             │ legitimate content                      │
```

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├──────────────┼──────────────────────────────────────────────┤
| SUSPICIOUS   | Minor inconsistencies, unknown sender,       |
|     ⚠️        | needs verification before acting             |
├──────────────┼──────────────────────────────────────────────┤
| PHISHING 🔴  | Fake domain, malicious links,                |
|              | attempts to steal data                        |
└──────────────┴──────────────────────────────────────────────┘
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Prevention Guidelines

Follow these best practices to stay safe:

1. ✅ **Verify the sender** — always check the full email address, not just the display name
2. 🔗 **Hover before clicking** — preview the real URL destination before opening any link
3. 📎 **Don't download unknown attachments** — even from seemingly familiar senders
4. 🔐 **Never share passwords or OTPs** — legitimate services will never ask for these via email
5. 🛡️ **Enable two-factor authentication (2FA)** — adds an extra layer of protection
6. 📣 **Report suspicious emails** — notify your IT/security team immediately

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Conclusion

Phishing remains a major cybersecurity threat due to human error and lack of awareness. This project demonstrates how phishing emails can be analyzed and detected using simple techniques.

By identifying key indicators and educating users, organizations can significantly reduce the risk of phishing attacks. **Awareness and vigilance are the most effective defenses against such threats.**

References

- Kaggle Phishing Email Dataset
- SpamAssassin Public Dataset
- Google Admin Toolbox - Message Header Analyzer
- MXToolbox Email Header Analyzer

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